STUDENT PROJECT GUIDE

Telco Customer Churn

Retention Management

Context

You are working with the Telco Customer Churn dataset: 7,043 customers, all in California, United States, in a single Q3 snapshot. The files describe demographics, location, services, billing, and customer status. They do not supply a longitudinal history or a calendar year. You report to the Head of Customer Retention.

Customer churn is costly because retention resources are limited and should be directed where they can change an outcome. The Head of Customer Retention needs a system that helps decide which existing customers should receive a retention intervention, and how that decision should be prioritized. A churn score on its own is not the goal.

Data provided

Use the course files supplied with this guide as the project dataset. The complete source field dictionary is incorporated below.

- **Demographics** — one row per `Customer ID`, with gender, age, senior-citizen status, marital status, dependents, and number of dependents. `Count` is a reporting field, not a customer characteristic.
- **Location** — one row per `Customer ID`, with country, state, city, ZIP code, and geographic coordinates.
- **Population** — one row per ZIP-code record, with an area-level population estimate. It joins to customers through `Zip Code`, not `Customer ID`.
- Services — one row per Customer ID for the documented fiscal-quarter snapshot, with referrals, tenure, accepted offer, subscribed services, contract, billing and payment information, monthly charge, and cumulative charges. Examples of coded values include Contract = Month-to-Month, One Year, or Two Year; Internet Service = Yes or No; Internet Type = None, DSL, Fiber Optic, or Cable; and Offer = the literal text None or Offer A–E. Preserve None when importing Excel (for example, keep_default_na=False); it is not an empty cell in the supplied Services file.
- **Status** — one row per `Customer ID` for the documented fiscal-quarter snapshot, with satisfaction, customer status, churn outcome, an existing churn score, predicted customer lifetime value, and churn reasons for customers who left.

`Demographics`, `Location`, `Services`, and `Status` join on `Customer ID`. `Population` joins through `Zip Code`. After joining, the intended analytical grain remains one row per customer; verify key uniqueness and row counts rather than assuming it.

Several fields are cumulative or model-derived. Tenure in Months, Total Charges, Total Refunds, Total Extra

Core task

Using the supplied **end-of-quarter customer snapshot**, classify existing customers as `Churned` versus `Stayed` and use the estimated churn risk to prioritize customers for a retention intervention.

Population: customers with `Customer Status` equal to `Churned` or `Stayed`; exclude `Joined` from the core model. **Target:** churned vs stayed. **Evaluation:** customer-level stratified holdout from the supplied snapshot.

Data dictionary

Demographics

Field	Meaning
`Customer ID`	Unique customer identifier.
`Count`	Reporting/dashboard field used to count customers in filtered views.
`Gender`	Customer gender: Male or Female.
`Age`	Customer age in years at fiscal-quarter end.

Field	Meaning
Under 30	Yes when Age < 30, otherwise No. Exact derivation of Age in all supplied rows.
Senior Citizen	Whether the customer is age 65 or older: Yes or No.
Married	Whether the customer is married: Yes or No.
Dependents	Yes when Number of Dependents > 0, otherwise No; an exact derivation in this sample.
Number of Dependents	Number of dependents living with the customer.

Location

Field	Meaning
Customer ID	Unique customer identifier.
Count	Reporting/dashboard field used to count customers in filtered views.
Country	Constant: United States for every customer. Not an informative predictor.
State	Constant: California for every customer. The sample does not cover multiple states.
City	City of the customer's primary residence.
Zip Code	ZIP code of the customer's primary residence.
Lat Long	Text representation of latitude and longitude; duplicates the information in the separate coordinate columns.
Latitude	Latitude of the customer's primary residence.
Longitude	Longitude of the customer's primary residence.

Population

Field	Meaning
ID	Unique row identifier for the population table.
Zip Code	ZIP code used to join population data to customer location.
Population	Population estimate for the ZIP-code area.

Services

Field	Meaning
Customer ID	Unique customer identifier.
Count	Reporting/dashboard field used to count customers in filtered views.
Quarter	Constant Q3 in this supplied snapshot; no year or sequence of quarters is present.
Referred a Friend	Yes when Number of Referrals > 0, otherwise No; an exact derivation in this sample.
Number of Referrals	Number of referrals made by the customer to date.
Tenure in Months	Months the customer has been with the company by quarter end.
Offer	Most recent accepted offer code: literal None (3,877 rows) or Offer A, B, C, D, E. None represents no offer in this field; preserve it rather than importing it as missing.
Phone Service	Whether the customer subscribes to home phone service: Yes or No.
Avg Monthly Long Distance Charges	Average monthly long-distance charges calculated through quarter end.

Field	Meaning
Multiple Lines	Whether the customer subscribes to multiple telephone lines: Yes or No.
Internet Service	Service subscription flag: Yes or No. This is not the connection technology.
Internet Type	Connection category: Fiber Optic, DSL, Cable, or literal None. None occurs for all 1,526 customers with Internet Service = No. Preserve it on import.
Avg Monthly GB Download	Average monthly download volume in GB through quarter end.
Online Security	Whether the customer subscribes to additional online-security service: Yes or No.
Online Backup	Whether the customer subscribes to additional online-backup service: Yes or No.
Device Protection Plan	Whether the customer subscribes to a device-protection plan: Yes or No.
Premium Tech Support	Whether the customer subscribes to premium technical support: Yes or No.
Streaming TV	Whether the customer uses the internet service for third-party TV streaming: Yes or No.
Streaming Movies	Whether the customer uses the internet service for third-party movie streaming: Yes or No.
Streaming Music	Whether the customer uses the internet service for third-party music streaming: Yes or No.
Unlimited Data	Whether the customer pays for unlimited data: Yes or No.
Contract	Current contract type: Month-to-Month, One Year, or Two Year.
Paperless Billing	Whether the customer uses paperless billing: Yes or No.
Payment Method	Payment method: Bank Withdrawal, Credit Card, or Mailed Check.
Monthly Charge	Current total monthly charge for all services.
Total Charges	Cumulative charges through quarter end.
Total Refunds	Cumulative refunds through quarter end.
Total Extra Data Charges	Cumulative extra-data charges through quarter end.
Total Long Distance Charges	Cumulative long-distance charges above plan allowance through quarter end.
Total Revenue	Cumulative monetary measure through quarter end. In every supplied row: Total Charges minus Total Refunds plus Total Extra Data Charges plus Total Long Distance Charges.

Status

Field	Meaning
Customer ID	Unique customer identifier.
Count	Reporting/dashboard field used to count customers in filtered views.
Quarter	Constant Q3 in this supplied snapshot; no year or sequence of quarters is present.
Satisfaction Score	Overall customer satisfaction rating from 1 (Very Unsatisfied) to 5 (Very Satisfied).
Customer Status	Quarter-end status: Churned, Stayed, or Joined.
Churn Label	Yes if the customer left during the quarter; No otherwise. Direct representation of the churn outcome.
Churn Value	1 if the customer left during the quarter; 0 otherwise. Numeric representation of the churn outcome.
Churn Score	0-100 churn score produced by an existing IBM SPSS Modeler predictive model; higher means greater predicted churn risk.
CLTV	Predicted customer lifetime value produced from corporate formulas and existing data.
Churn Category	High-level reason for churn: Attitude, Competitor, Dissatisfaction, Other, or Price; collected for customers who left.
Churn Reason	Specific reason given by a customer who left; directly related to Churn Category.

Field	Meaning
Modeling note: Customer Status, Churn Label and Churn Value encode the same outcome. Churn Category and Churn Reason are post-churn fields and are blank for the 5,174 customers who did not churn. Age generates Under 30 and Senior Citizen; Number of Dependents generates Dependents; Number of Referrals generates Referred a Friend. These are derived representations, not independent measurements.	

Provenance

The sources below document dataset provenance.

- **IBM Cognos Analytics — Telco Customer Churn sample:** https://www.ibm.com/docs/en/cognos-analytics/12.0.x?topic=samples-telco-customer-churn